



Instructor: Date:

Learner: Regional Branch:

Program: Center:

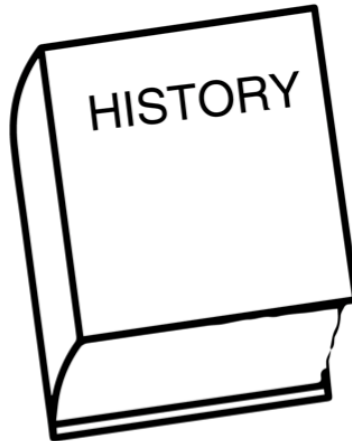
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Let's Start!

Systems Theory Timeline



Learning Outcome:

Understands systems theories, business processes and basic concepts on IT.
Create a map of processes that allows identifying areas in an information system, using IT tools to generate reports based on the company's needs.

Ask your partner:

- Do you know what a social organization is? Can you explain?
- Can you explain some of the differences between software applications and computer programs?

Can Do Statements:

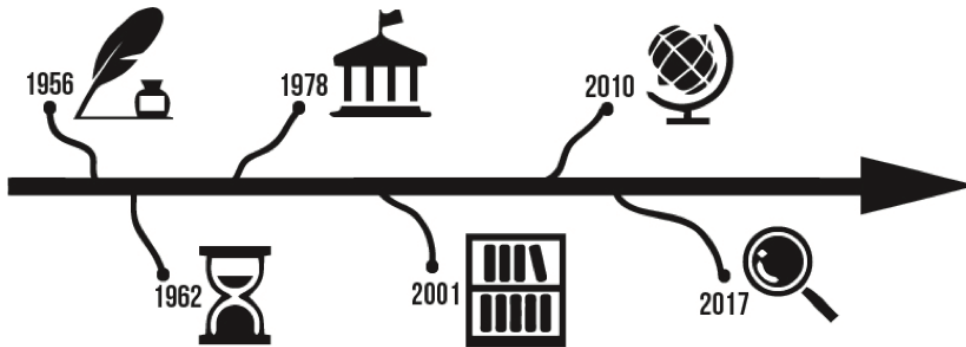
- I can write a list of important information related with systems theory history.
- I can answer basic questions related to systems theory history.
- I can talk about systems theory development within software and computing.





Let's Explore!

Check systems history and learn about phenomena interactions!



Instructions: Read the text below.

Larry: Carol, I need your help!

Carol: How can I help you?

Larry: Well, I want to know a little bit about systems theory in information systems. I'm sure you know a lot about that.

Carol: Well, we need to understand two aspects first. One of them is functional, where we record, store, and distribute algorithmic expressions, and the other one is structural, where we assess people, processes, data, models, and language to design a structure with a purpose or function for an organization

Larry: That's new to me! Can you tell me about the purpose of systems theory?

Carol: Sure! The main purpose is to make good decisions and to maintain control of an organization or of a company. It helps workers and managers analyze problems and create new products.

Larry: It sounds interesting. What implications does systems theory have for a company?

Carol: It's important for all companies. To understand that, we need to talk about input, processing, and output. Input is about data capture, processing is about data applications, and output is about the transference of the results that the applications show to the managers who make decisions based on this information. Feedback is next.

Larry: I have another question. What do you know about the history of systems theory?

Carol: Well, the Society for General Systems Research, born in 1956, studies all phenomena in any system and identifies common patterns in a variety of systems in search of a unified theory.

Larry: I understand everything, but the history of systems theory in software and computing development is not clear to me.

Carol: Well, when the model of electrical circuits and other systems appeared, the new software applications and computing programs originated too. All of these developments demonstrate practical applications of systems theory.

Larry: You're very kind, Carol. Lunch is on me!

**Let's Practice!****Basic Systems Theory Information**

Instructions: Select the correct answer.

1. What is data?
 - a. It is a process in the company.
 - b. It is a decision you make.
 - c. It is a collection of information.
2. What is the function of input?
 - a. It captures information.
 - b. It transfers information.
 - c. It changes information.
3. What is the function of output?
 - a. It transfers information.
 - b. It changes information.
 - c. It captures information.

**Let's Practice!****History and Purposes of Systems Theory**

Instructions: Classify the sentences by writing the number of the sentence in the correct category.

The Purpose of System Theory	History of Systems Theory

1. The Society for General Systems Research studies all phenomena in any system and identifies common patterns in a variety of systems.
2. It helps to have control and make good decisions.
3. It is a fact that when electrical circuits and other systems appeared, the new software applications and computing programs originated too.
4. It analyzes problems and provides feedback.





Extension Activity

Make a brochure about system theory history.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about system theory history.
- Summarize the main events and write them in the brochure.
- Write about the importance of the history of system theory.
- Share your brochure with the class.

Self-Assessment

Yes

No

- I find information about system theory history. ☐ ☐
- I summarize important historical information. ☐ ☐
- I identify what the importance of system history is. ☐ ☐



Target Words

- **feedback** (noun): information for improvement
e.g. Feedback is very important for the company.
- **data** (noun): information; statistics
e.g. Processing extracts important patterns from the data.
- **basic** (noun): relating to something fundamental
e.g. I have some basic questions about the history of systems theory.
- **view** (noun): perspective
e.g. It has two points of view.
- **common** (noun): something usual, frequent, or average
e.g. Identifying common patterns in a variety of systems for a unified theory.
- **capture** (verb): to obtain
e.g. The input captures data.



Learning Points

Simple Connectors

You use these words to connect ideas and sentences. They help you provide more information.

The most common connectors are **and**, **but**, **or**, and **so**.

You use **and** to join two sentences or ideas and to add more information. For example:

- *Systems theorist research about life systems **and** use the information to solve problems.*

You use **or** to mention possibilities and give options. For example:

- *A system consists of attributes **or** qualities.*

You use **but** to contrast information, and to give surprising or different information. For example:

- *Systems theorists consider philosophy theories **but**, does it take into account science?*

You use **so** to talk about consequences and results. For example:

- *Systems theory is an interdisciplinary study, so it has many different perspectives.*

Examples

- *We record, store, and distribute algorithmic expressions.*
- *Thank you Carol, but I have another question. What do you know about the history of systems theory?*
- *As the model of electrical circuits appeared along with other systems, so did new software applications and computing programs.*





Learning Points

Simple Present Tense - Affirmative Form

You use the **simple present** to talk about general habits, present facts, and repetitive actions.

In **simple present**, you use the base form of the verb to make an affirmative statement. For example:

- ✓ *Theorists **work** with philosophers to understand systems.*
- ✗ *Theorists to work with philosophers to understand systems.*

Use just the base form to make affirmative statements. Also, remember the structure:

Subject + verb in base form + complement
*I **study** computing and software matters.*
*I **think** theorists need to emphasize in systems functions.*

Take into account that when you make an affirmative statement and you are using a third person (he, she, it), you have to add S at the end of the main verb.

If the verb in your sentence ends in s, z, ch, sh, or x, you add ES.

Examples:

fix: fixes - catch: catches - finish: finishes

If the verb ends in consonant + y, you remove the Y and add IES.

Examples:

- study: studies - imply: implies - try: tries

If the verbs ends in vowel + y, you just add S at the end of it.

Examples:

-play: plays - stay: stays

There are some irregular verbs that don't require a specific rule to form the conjugation of the third person. Those verbs are:

-have: has -go: goes -do: does

Keep in mind that the third person can be the personal pronouns he, she, or it, or a noun like a place, a person, or an object. For example:

*Caroline **plays** with different computer programs.*
*She **plays** with different computer programs.*

Examples

- *We record, store, and distribute algorithmic expressions.*
- *I need to write about the history of systems theory.*

